

ADVANCED SQL NULL FUNCTIONS

EXERCISE B

Q1. SELECT employee_id,
name,
IFNULL(salary, 0) AS salary_with_default
FROM Employees;

Employees

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2. SELECT employee_id,
name,
IFNULL(department, 'Not Assigned') AS department_name
FROM Employees;

Employees

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3. SELECT order_id,
customer_id
FROM Orders
WHERE customer_id IS NULL;

Orders

order_id	customer_id
103	NULL

Q4. SELECT order_id,
customer_id,
IFNULL (delivery_date, 'Pending') AS delivery_status
FROM Orders;

Orders

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5. SELECT student_id,
name,
IFNULL (grade, 0) AS final_grade
FROM Students;

Students

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6. SELECT COUNT (Student-id) AS not_graded_count
FROM Students
WHERE grade IS NULL;

not_graded_count
1

Q7. SELECT Product-id,
name,
price - COALESCE (discount, 0) AS final-price
FROM Products;

Products

product-id	name	price	final-price
501	Keyboard	25	0
502	Mouse	15	5
503	Monitor	100	10.0

Q8. SELECT COUNT (*) AS missing_email_count
FROM Customers
WHERE email IS NULL;

OR

SELECT COUNT (customer-id) AS missing_email_count
FROM Customers
WHERE email IS NULL;

Customers

customer-id

name

missing_email_count
2

5
2
2
3

Brandon
John
Joseph
Alice

Q9. SELECT customer_id,
name,
IFNULL(email, 'No email') AS email_display
FROM Customers;

Customers

customer_id	name	email_display
1	Linda	No email
2	Joseph	Joseph@mail.com
3	Nia	No email

Q10. SELECT payment_id, status,
IFNULL(method, 'Unknown') AS method_display
FROM Payments;

Payments

payment_id	method_display	status
301	Credit	NULL
302	Pay Pal	Success
303	Unknown	Failed

Q11. SELECT item_id,
item_name,
IFNULL(quantity, 0) AS quantity_checked
FROM Inventory;

Inventory

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Erasor	0

Q12 SELECT emp_id,
COALESCE ('bonus', 'commission') AS first_available_reward
FROM Employees_Extra;

Employees_Extra

emp_id	first_available_reward
1	300
2	100

Q13 SELECT COUNT(*) AS no-room-count
FROM Classes
WHERE room IS NULL;

OR

SELECT COUNT (class_id) AS no_room-count
FROM Classes
WHERE room IS NULL;

no-room-count
2

Q14 SELECT student_id,
date,
IFNULL (status, 'Not Marked') AS attendance_status
FROM Attendance;

Attendance

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

①5. SELECT account-id,
 COALESCE (account-type, 'Unknown') AS type-display,
 COALESCE (balance, 0) AS balance-checked
 FROM Bank_Accounts;

Bank_Accounts

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

①6. SELECT project-id,
 title,
 IFNULL (start-date, 'TBD') AS start-display
 FROM Projects;

Projects

project-id	title	start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

①7. SELECT review-id,
 product-id,
 COALESCE (comment, 'No comment') AS comment-display,
 COALESCE (rating, 0) AS rating-display
 FROM Reviews;

Reviews

review_id	product_id	comment	rating
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

① 18. SELECT supplier_id,
name,
COALESCE(phone, alt_phone, 'No contact') AS contact_number
FROM Suppliers;

Suppliers

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No contact

① 19. SELECT user_id,
COALESCE(theme, 'Light') AS theme_set,
COALESCE(language, 'English') AS language_set,
COALESCE(timezone, 'UTC') AS timezone_set
FROM User_Settings;

User - Settings

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q20. SELECT record_id,
 machine_id,
 COALESCE(issue, 'Unknown Issue') AS issue_log,
 COALESCE(technician, 'Not Assigned') AS technician_name
 FROM Maintenance;

Maintenance

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex